CariSurg MedTech Pathways Programme

# Week 5  Feasibility Memo

To: Dr. L. De Freitas & Mercer General Hospital ED Board

From: Mithra Ramoutar, Clinical AI Research Trainee

Re: Is the Yale EMLC triage dataset good enough to build a triage model from?

## 1. Verdict

**This dataset is suitable for building a baseline triage model, provided that two post-visit columns are excluded to prevent artificial inflation of results, temperature thresholds are applied in Fahrenheit rather than Celsius, and any performance figures are understood as benchmarks for a US academic hospital  not guarantees of what the tool will achieve at Mercer General.**

## 2. Dataset Summary

The Yale Emergency Medicine Learning Collaborative (EMLC) triage dataset contains 55,121 patient encounters drawn from a US academic hospital ED. Each encounter includes patient demographics (age, gender, race, ethnicity, insurance status, arrival mode), seven triage vital signs (heart rate, blood pressure, respiratory rate, oxygen saturation, temperature, and glucose), and 200 binary flags recording whether each of 200 chief complaints was documented at triage. The target variable is the Emergency Severity Index (ESI)  a five-level scale where ESI 1 represents the most critically ill patient.

| ESI Level | Description | Encounters | % of Total |
|---|---|---|---|
| 1  Immediate | Life-threatening; resuscitation required | 77 | 0.1% |
| 2  Emergent | High risk; requires rapid assessment | 17,924 | 32.5% |
| 3  Urgent | Stable but needs multiple resources | 27,010 | 49.0% |
| 4  Less Urgent | Needs one resource; not time-sensitive | 8,896 | 16.1% |
| 5  Non-Urgent | Minimal resources; could wait | 1,214 | 2.2% |

The patient population is adult-only (age 18–107, median 55 years) and skews older at higher acuity levels  ESI 1 patients have a mean age of 69 years, ESI 5 patients 46 years. Female patients account for 58% of encounters. White or Caucasian patients represent 53% of the dataset; Black or African American patients 29%. Medicaid and Medicare together cover 71% of encounters, suggesting a publicly insured, lower-income population.

## 3. Top Three Data Quality Concerns

**Concern 1: Severe class imbalance  very few critically ill patients** ESI 1 accounts for only 77 encounters out of 55,121  less than 0.2% of the dataset. A triage model trained on this data could score very highly by simply predicting ESI 3 for everyone, while completely failing to identify the patients who most urgently need help. This is the most important quality concern for patient safety. Mitigation: the model will be evaluated specifically on how often it catches ESI 1 and ESI 2 patients, not on overall accuracy. Class-weighting techniques will be applied during training.

### Concern 2: Two post-visit columns would give the model information it cannot have at triage

The columns 'disposition' (whether a patient was admitted or discharged) and 'previousdispo' (outcome of prior ED visits) are only known after the visit is complete. Using them as model inputs would give the algorithm the answer before the question is asked, inflating performance figures in ways that would not replicate in live use. Both columns will be excluded from all modelling. This is not a flaw in the dataset  it is a modelling decision that must be enforced explicitly.

### Concern 3: US academic hospital data may not reflect Mercer General's patient population

This dataset comes from a single US academic hospital. Mercer General Hospital serves a Trinidadian population with a different demographic profile, disease burden, resource availability, and documentation culture. A model that performs well on this data may perform substantially differently when deployed locally. This is the same distribution shift risk documented in the Epic Sepsis Model case. Mitigation: any performance figures from this dataset must be treated as development benchmarks, not deployment targets. Local validation on Mercer ED data is required before go-live.

## 4. Top Three Reasons to Proceed

### Reason 1: A real, complete triage label exists for every encounter

Every one of the 55,121 encounters carries a genuine ESI level assigned by a triage nurse. There are no missing target values. This is the foundation of any supervised learning model  without a reliable label, no amount of data quality fixes the problem. The ESI scale is the same acuity framework used internationally and compatible with the CTAS system at Mercer.

**Reason 2: Triage vital signs are complete and clinically meaningful** All seven vital sign columns  heart rate, systolic and diastolic blood pressure, respiratory rate, oxygen saturation, temperature, and glucose  are fully recorded with no missing values. The clinical validity checks confirm expected patterns: ESI 1–2 patients have lower oxygen saturation and higher heart rates than ESI 4–5 patients. The data behaves as a clinician would expect, which is the most important signal that it is trustworthy enough to model on.

### Reason 3: 55,000 encounters provides enough volume to train a stable baseline model

Even after excluding leakage columns and applying class-weighting for the rare ESI 1 cases, 55,121 encounters is a substantial training set. It is large enough to support train-test splits, cross-validation, and subgroup fairness analysis across race and insurance status categories. The 200 chief complaint flags, while sparse individually, collectively capture the full range of presenting conditions.

## 5. Caveats

• Temperature is recorded in degrees Fahrenheit. Clinical fever threshold (>100.4°F) and hypothermia threshold (<95°F) differ from Celsius-based guidelines and must be applied correctly in feature engineering.

• Chief complaint flags are sparse binary columns  200 in total, but most individual flags are present in fewer than 500 encounters. Feature selection will be required to avoid introducing noise into the model.

• 164 glucose values exceed 600 mg/dL (maximum 1,066 mg/dL). These are physiologically possible in diabetic emergencies but may include data entry errors. They will be reviewed against ESI level and either retained with documentation or capped at a defensible threshold.

• Correlations between individual features and ESI are modest (highest is O2 saturation at 0.178). This is expected  triage acuity is determined by patterns across multiple features, not by any single measurement. A model combining features will outperform individual correlations.

• External validity for a Caribbean ED is untested. All performance figures from this dataset are US hospital benchmarks and should not be presented to the ED Board as expected Mercer performance without a local validation study. ..

## Data Quality Dashboard

The eight-panel dashboard below summarises the key exploration findings. Panels show ESI distribution, patient race, heart rate and O2 saturation by acuity level, top 15 chief complaints, age distribution, insurance status, and glucose outlier profile.

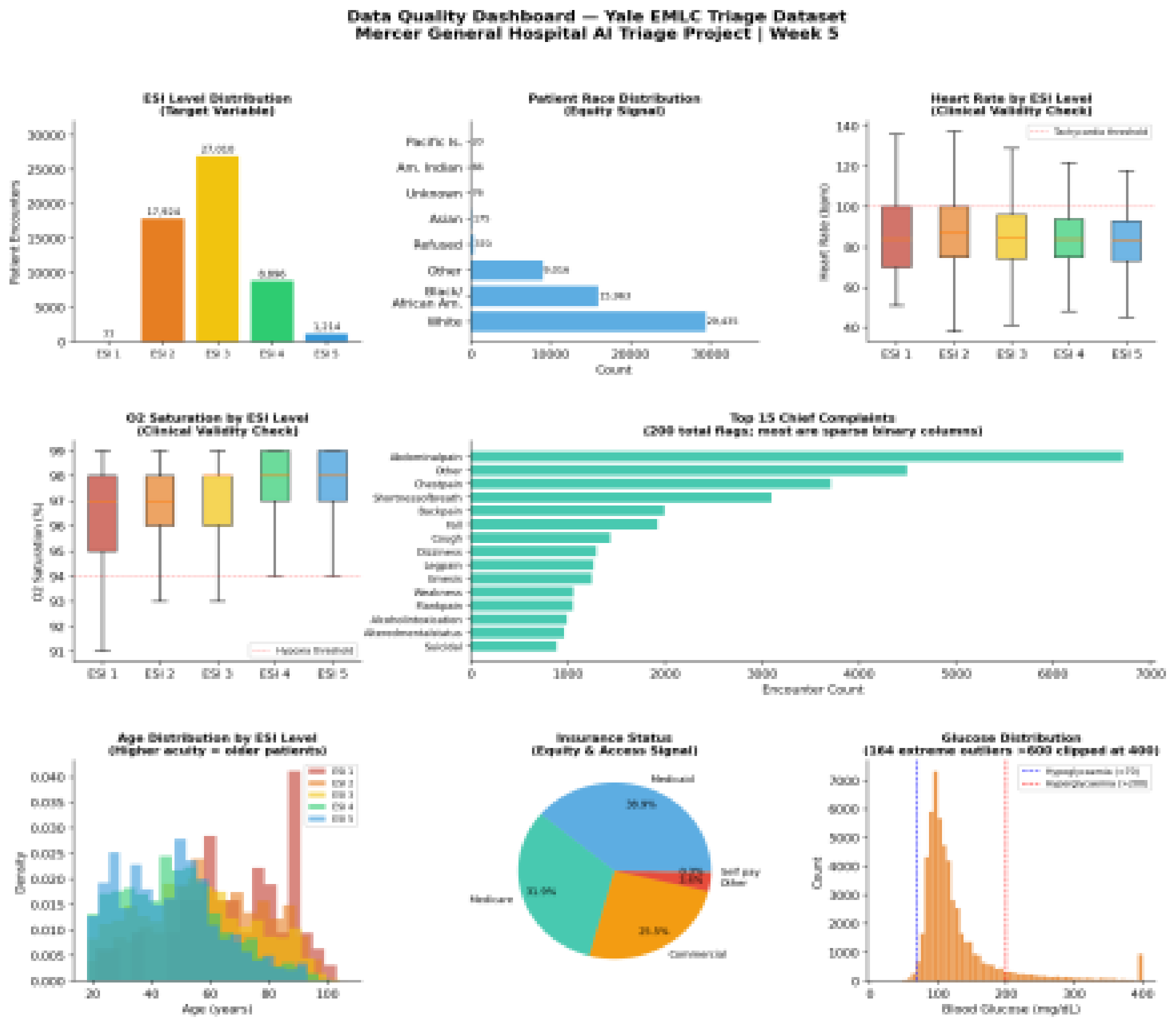


Figure 1. Data quality dashboard Yale EMLC triage dataset, 55,121 encounters, 226 features.

## Top 10 Feature Shortlist

The following features are ranked by a combination of clinical intuition and simple correlation with ESI level. Leakage columns (disposition, previousdispo) are explicitly excluded. These are presented as hypotheses to test in Week 6 modelling  not proven predictors.

| Rank | Feature | Clinical Rationale | Correlation w/ ESI |
|---|---|---|---|
| 1 | triage_vital_o2 | Oxygen saturation directly reflects respiratory compromise the single strongest physiological predictor of acuity in this dataset. | +0.178 (highest of all vitals) |
| 2 | triage_vital_hr | Tachycardia (HR >100) is a marker of physiological stress across sepsis, haemorrhage, and pain. Higher urgency cases show elevated HR. | -0.095 |
| 3 | triage_vital_rr | Respiratory rate is a sensitive early warning sign  elevated RR often precedes clinical deterioration before other vitals change. | -0.095 |
| 4 | triage_glucose | Hyperglycaemia and hypoglycaemia both flag metabolic crisis. Extreme glucose values are disproportionately present at ESI 1–2. | -0.078 |
| 5 | age | Older patients are significantly over-represented at ESI 1–2 (mean age 69 and 61 respectively vs. 46 at ESI 5), capturing frailty and comorbidity burden. | Clinical gradient confirmed |
| 6 | cc_shortnessofbreath | Shortness of breath is the second most clinically urgent complaint, strongly associated with cardiac and respiratory emergencies at ESI 1–2. | High ESI 1–2 co-occurrence |
| 7 | cc_chestpain | Chest pain is the most common presenting complaint linked to cardiac events; third-highest volume complaint at 3,712 cases. | High ESI 2 co-occurrence |
| 8 | cc_alteredmentalstatus | Altered mental status signals neurological, metabolic, or toxic emergencies and is consistently associated with higher acuity assignments. | High ESI 1–2 co-occurrence |
| 9 | triage_vital_sbp | Systolic blood pressure extremes (hypotension <90 or hypertensive crisis >200) are red flags; mid-range values show less discrimination. | Low overall (-0.001) |
| 10 | arrivalmode | Ambulance arrival is a proxy for pre-hospital acuity assessment ambulance patients are disproportionately triaged ESI 1–3. | Categorical clinical signal confirmed |

## Cleaning Decisions Log

| Decision | Column(s) | Rule | Rationale |
|---|---|---|---|
| Exclude leakage columns | disposition, previousdis po | Remove from all model inputs | Known only post-visit; would inflate metrics |
| Retain glucose outliers (flagged) | triage_glucose | Flag values >600; review vs. ESI before modelling | Possible diabetic emergencies; not auto-removed |
| Temperature unit noted | triage_vital_temp | All thresholds applied in °F not °C | Dataset uses Fahrenheit throughout |
| Unnamed index column dropped | Unnamed: 0 | Remove  row index only | No clinical information content |
| Chief complaint sparsity noted | All cc_ columns | Apply feature selection in Week 6 | 200 flags; most add minimal signal individually |